Learning Nonlinear Relationships with a Multilayer Perceptron (MLP)

Fundamentals of AI | Individual or pair work | Suggested duration: 10 days

MAIN SCORE: 100 points PRACTICAL WORK: 60 points THEORY: 40 points

Scoring: The main project is worth 100 points: 60 for implementation and practical analysis, and 40 for MLP theory questions embedded throughout the coding sections. A separate bonus task after Section 7 is worth up to 10 additional points and is outside the main score.

Project objective

You will work with a shared synthetic master dataset containing 1,000 observations. At the beginning of the starter notebook, your student ID (or both IDs for a pair) is used with a reproducible seed to select a group-specific subset of 100 observations. Each selected observation has three inputs, x, y, and z, and one output, u.

The exact relationship between the inputs and output is not provided. Your task is to explore the data, build an MLP, test several hyperparameter configurations, and draw a reasoned conclusion about the likely relationships among the variables.

The data contain noise, so do not expect zero prediction error. The goal is to perform well on unseen data, not to memorize the training set.

Key information and rules

- Your group-specific dataset must contain exactly 100 rows with columns x, y, z, and u. The shared master CSV contains 1,000 rows; do not edit it.
- Run the starter notebook to create your 100-row CSV subset using your student ID number(s). Keep the notebook and master CSV together, or upload the master CSV when prompted in Google Colab.
- The group seed uses the last four digits of each student ID. For a pair, sort the two four-digit suffixes from smallest to largest and concatenate them. For example, 1234 and 5678 produce seed 12345678.
- Use the same seed for subset selection and the Training/Validation/Test split, so your experiment is reproducible.
- Each network configuration may be trained for at most 100 epochs. Early Stopping is allowed.
- Use Validation data to select the final model. Evaluate on Test data only after selection; do not tune again based on Test results.
- AI tools may help with coding, but you must explain your code, parameter shapes, hyperparameters, plots, and conclusions.

Section 1 - Load and inspect the data | 8 points

Practical work - 5 points

- (1 point) Read the group CSV file and display its first five rows.
- (2 points) Report the number of rows and columns, column names, data types, and number of missing values.
- (1 point) Check for duplicate rows or unusual values, and briefly explain how you checked.
- (1 point) Record the group seed and data filename in your notebook.

Theory questions - 3 points

- Before splitting, write the shapes of input array X and output array u. Explain what each dimension represents, including the number of observations and input features.

Section 2 - Exploratory data analysis and visualization | 12 points

Practical work - 7 points

- (2 points) Create descriptive statistics for x, y, z, and u.
- (3 points) Draw scatter plots: u versus x, u versus y, and u versus z.
- (2 points) Give each plot an appropriate title, axis labels, and readable scales.

Theory questions - 5 points

- Using your plots, describe each trend as increasing, decreasing, or unclear. Discuss whether each possible relationship appears linear or nonlinear; explain the visual evidence for at least one.
- Explain why a correlation or visual trend does not, by itself, prove causation.

Section 3 - Data splitting and preprocessing | 12 points

Practical work - 6 points

- (3 points) Split the data into approximately 70% Training, 15% Validation, and 15% Test. Use your group seed.
- (3 points) Standardize the input features. Fit scaling parameters on Training data only. If you scale u, inverse-transform predictions before reporting errors in the original output scale.

Theory questions - 6 points

- Distinguish the roles of Training, Validation, and Test data, and explain why all three are needed.
- What is data leakage? Explain what can go wrong if the mean and standard deviation for standardization are calculated using the entire dataset, including Validation and Test.

Section 4 - Design hyperparameter experiments | 14 points

Practical work - 8 points

- (3 points) Test at least five different configurations.
- (3 points) Examine at least three hidden-layer structures and at least two activation functions. Also change the learning rate or batch size in at least one experiment.
- (2 points) Record architecture, activation, learning rate, batch size, epochs actually run, and one consistent Validation-error metric in a table.

Theory questions - 6 points

- Compare ReLU and tanh using your results. Explain how ReLU can help learning and when tanh may saturate and produce small gradients. Relate your conclusion to the data and observed results; neither activation is always best.
- For regression, why is a linear output activation usually used? Explain what limitation sigmoid or ReLU could impose on the range of predictions.

Section 5 - Train and select an MLP | 18 points

Practical work - 11 points

- (4 points) Build an MLP for regression with Keras. You do not need to define weights manually.
- (3 points) Train each configuration for no more than 100 epochs. If you use Early Stopping, retain the reason training stopped and the training-history plot.
- (4 points) Compare configurations using Validation results and justify your final model selection.

Theory questions - 7 points

- For your selected architecture, identify the input neurons, neurons in every hidden layer, and output neurons. Write each Dense layer kernel and bias shape, and calculate total trainable parameters. Your calculation must match your code.
- What role does the learning rate play when weights are updated by backpropagation? Explain how a rate that is too large or too small affects error reduction and training stability.

Section 6 - Final evaluation | 12 points

Practical work - 6 points

- (3 points) Only after selecting the model, calculate and report MAE and RMSE on the Test data.
- (3 points) Plot actual u values against predicted u values. Also show Training- and Validation-loss histories for the selected model.

Theory questions - 6 points

- Explain MAE, MSE, and RMSE. How do they differ in their treatment of large errors, and what are the units of MAE and RMSE relative to u?
- What does R-squared represent? Interpret values near 1, near 0, and below 0.
- Compare the computational cost of MAE, MSE, RMSE, and R-squared. Consider their time complexity for n observations and explain why practical speed differences are usually not decisive here.

Section 7 - Infer the relationships among variables | 16 points

Practical work - 10 points

- (2 points) Using plots and predictions, describe the relationship between u and x and discuss evidence of nonlinearity.
- (2 points) Describe the relationship between u and y. Is it similar to the relationship with x, or does it differ?
- (2 points) Describe the relationship between z and u, and argue whether it appears approximately linear.
- (4 points) Write a hypothesis about the overall relationship between u and x, y, and z. Explain how noise prevents a perfect match.

Theory questions - 6 points

- Examine Training- and Validation-loss curves. What patterns may indicate underfitting or overfitting? Suggest a corrective action.
- Why should you not repeatedly use Test results to adjust architecture or hyperparameters after selecting a model?
- Why is low Training error alone not enough to choose a model?

Bonus task - Controlled investigation of a relationship | Up to 10 extra points

This task is separate from the 100-point main score.

- (4 points) Hold two inputs constant and vary the third over a range, preferably within the Training-data range.
- (3 points) Calculate predicted u values and plot predicted output against the varying input. Add a title and axis labels.
- (3 points) Analyze the trend. If justified, propose an approximate nonlinear or power-law relationship and discuss the limitations of your conclusion.

Section 8 - Report and presentation | 8 points

Practical work - 7 points

- (4 points) Prepare a 5-7 minute presentation covering the data, key plots, main experiments, final-model selection, and conclusions.
- (3 points) Submit a tidy, runnable notebook with code, plots, and a comparison table for at least five configurations.

Theory question - 1 point

- During the presentation, answer a short question about weight shapes, learning rate, activations, evaluation metrics, or Training versus Test data.

Scoring summary

Theory points are integrated into the sections; they should not be answered as a separate block detached from the coding work. The bonus task is worth up to 10 additional points, outside the 100-point main score.

Part	Topic	Practical	Theory	Total
1	Load and inspect data	5	3	8
2	Exploratory analysis and visualization	7	5	12
3	Data splitting and preprocessing	6	6	12
4	Hyperparameter experiments	8	6	14
5	Train and select an MLP	11	7	18
6	Final evaluation	6	6	12
7	Infer variable relationships	10	6	16
8	Report and presentation	7	1	8
TOTAL	Main project	60	40	100

Reference - Dense-layer dimensions and parameter counts

If a Dense layer maps n input values to m neurons:

- Weight matrix: kernel.shape = (n, m).
- Bias vector: bias.shape = (m,).
- Trainable parameters: n x m + m.

Example: X_batch.shape = (batch_size, 3) and prediction.shape = (batch_size, 1). Total trainable parameters are the sum across all layers. The exact shape of u labels depends on how you prepare target data.

What to submit

- A Jupyter Notebook containing your code, tables, plots, and explanations.
- A comparison table covering at least five MLP configurations.
- A short presentation or in-person explanation of your results.

Final reminder: the best model is not necessarily the one with the lowest Training error. It should also perform well on Validation data, and final Test evaluation must happen without further tuning.